

DSA Daily Quiz 4

Total points 11/15 ?

The respondent's email (**omjbhaleraowork@gmail.com**) was recorded on submission of this form.

Full Name *

om jairam bhalerao

Last TWO Digits of PRN *

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✓ The number of edges from the root to the node is called _____ of the *1/1 tree.

☐ Height

☒ Depth

☐ Length

☐ Width



✓ The number of edges from the node to the deepest leaf is called _____ of the tree. *1/1

- ☒ Height
- ☐ Depth
- ☐ Length
- ☐ Width



✓ What is a full binary tree? * 1/1

- ☒ Each node has exactly zero or two children
- ☐ Each node has exactly two children
- ☐ All the leaves are at the same level
- ☐ Each node has exactly one or two children



✓ What is a complete binary tree? * 1/1

- ☐ Each node has exactly zero or two children
- ☐ A binary tree, which is completely filled, with the possible exception of the bottom level, which is filled from right to left
- ☒ A binary tree, which is completely filled, with the possible exception of the bottom level, which is filled from left to right
- ☐ A tree in which all nodes have degree 2



✓ Which of the following rotations is called a double rotation in AVL Tree? * 1/1

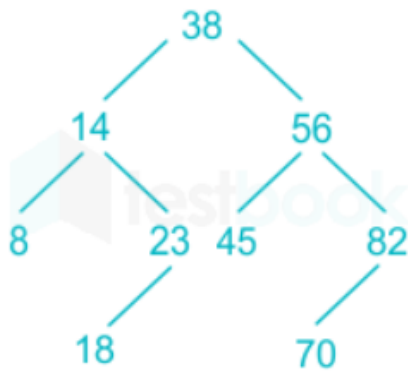
- ☐ RL and RR
- ☐ LR and LL
- ☐ LL and RR
- ☒ LR and RL



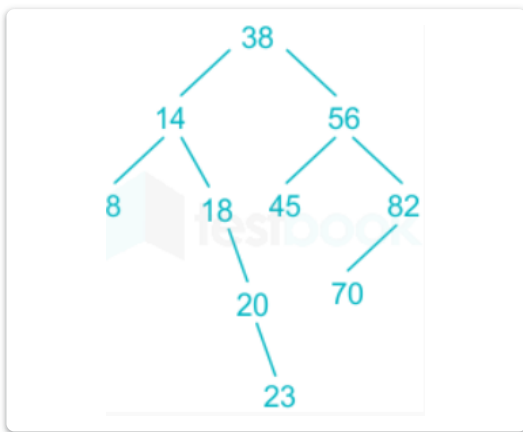


1/1

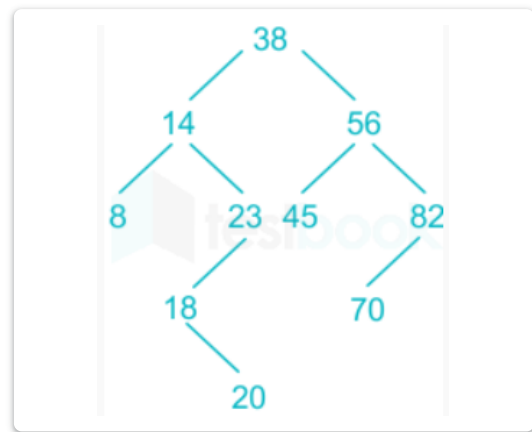
Consider the following AVL tree.



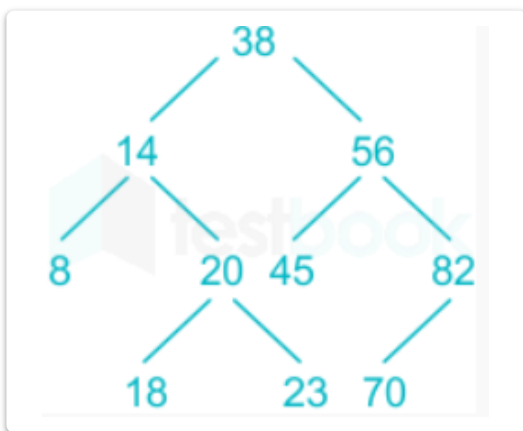
What will be the resultant AVL tree after insertion of node with value 20?



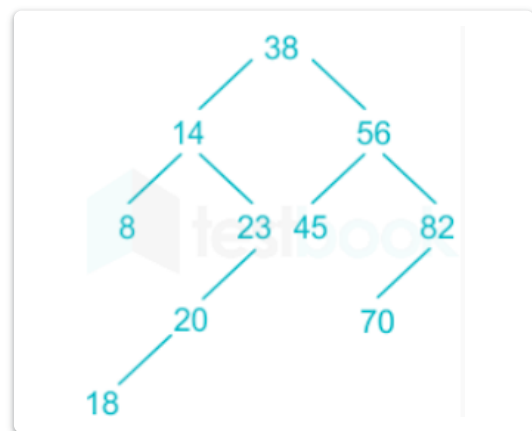
☐ Option 1



☐ Option 2



☒ Option 3



☐ Option 4



✗ What order should we insert the elements {1,2,...,7} into an empty AVL tree so that we don't have to perform any rotations on it? *0/1

- ☒ {4,2,1,6,3,5,7} ✗
- ☐ {4,2,6,1,3,5,7}
- ☐ {4,2,1,6,3,7,5}
- ☐ None of the above

Correct answer

- ☒ {4,2,6,1,3,5,7}

✗ A height balanced binary search tree is called_____ * 0/1

- ☐ height tree
- ☐ AVL tree
- ☐ binary tree
- ☒ binary search tree ✗

Correct answer

- ☒ AVL tree

✓ The balancing factor for AVL tree = * 1/1

- ☒ hL-hR ✓
- ☐ hR-hL
- ☐ hL-hL
- ☐ hR-hR



✗ What is the average case time complexity for finding the height of the binary tree? *0/1

☐ $h = O(\log \log n)$

☐ $h = O(n \log n)$

☒ $h = O(n)$

✗

☐ $h = O(\log n)$

Correct answer

☒ $h = O(\log n)$

✗ In a full binary tree if number of internal nodes is I , then number of leaves L are? *0/1

☐ $L = 2 * I$

☐ $L = I + 1$

☒ $L = I - 1$

✗

☐ $L = 2 * I - 1$

Correct answer

☒ $L = I + 1$



✓ In a full binary tree if number of internal nodes is I , then number of nodes N are? *1/1

- ☐ $N = 2 \cdot I$
- ☐ $N = I + 1$
- ☐ $N = I - 1$
- ☒ $N = 2 \cdot I + 1$



✓ In a full binary tree if there are L leaves, then total number of nodes N are? *1/1

- ☐ $N = 2 \cdot L$
- ☐ $N = L + 1$
- ☐ $N = L - 1$
- ☒ $N = 2 \cdot L - 1$



✓ Which of the following is incorrect with respect to binary trees? * 1/1

- ☐ Let T be a binary tree. For every $k \geq 0$, there are no more than 2^k nodes in level k
- ☐ Let T be a binary tree with λ levels. Then T has no more than $2\lambda - 1$ nodes
- ☐ Let T be a binary tree with N nodes. Then the number of levels is at least $\text{ceil}(\log(N + 1))$
- ☒ Let T be a binary tree with N nodes. Then the number of levels is at least $\text{floor}(\log(N + 1))$



✓ Suppose a binary tree is constructed with n nodes, such that each node has exactly either zero or two children. The maximum height of the tree will be? *

☐ $(n+1)/2$

☒ $(n-1)/2$



☐ $n/2 - 1$

☐ $(n+1)/2 - 1$

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